

Tech & Teach

Digital Chip Design Program

RTL Design

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Task#3

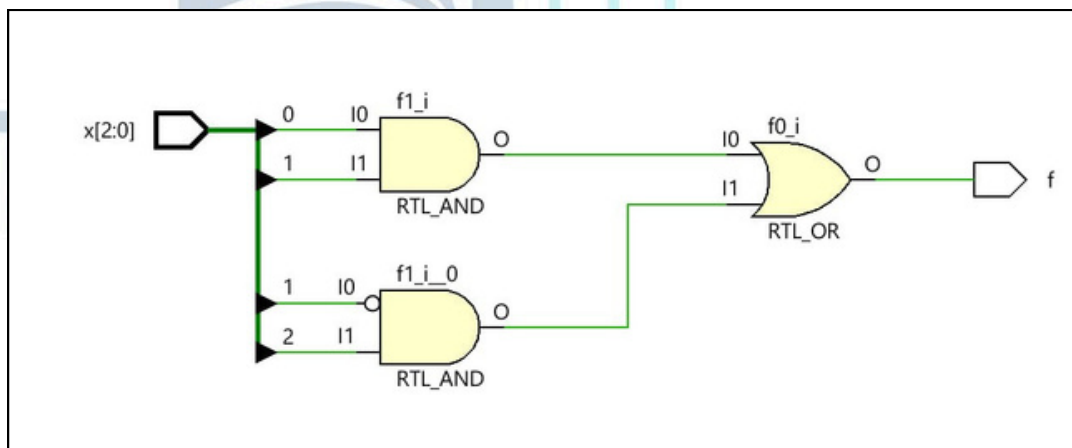
1/7/2026

Behavioral

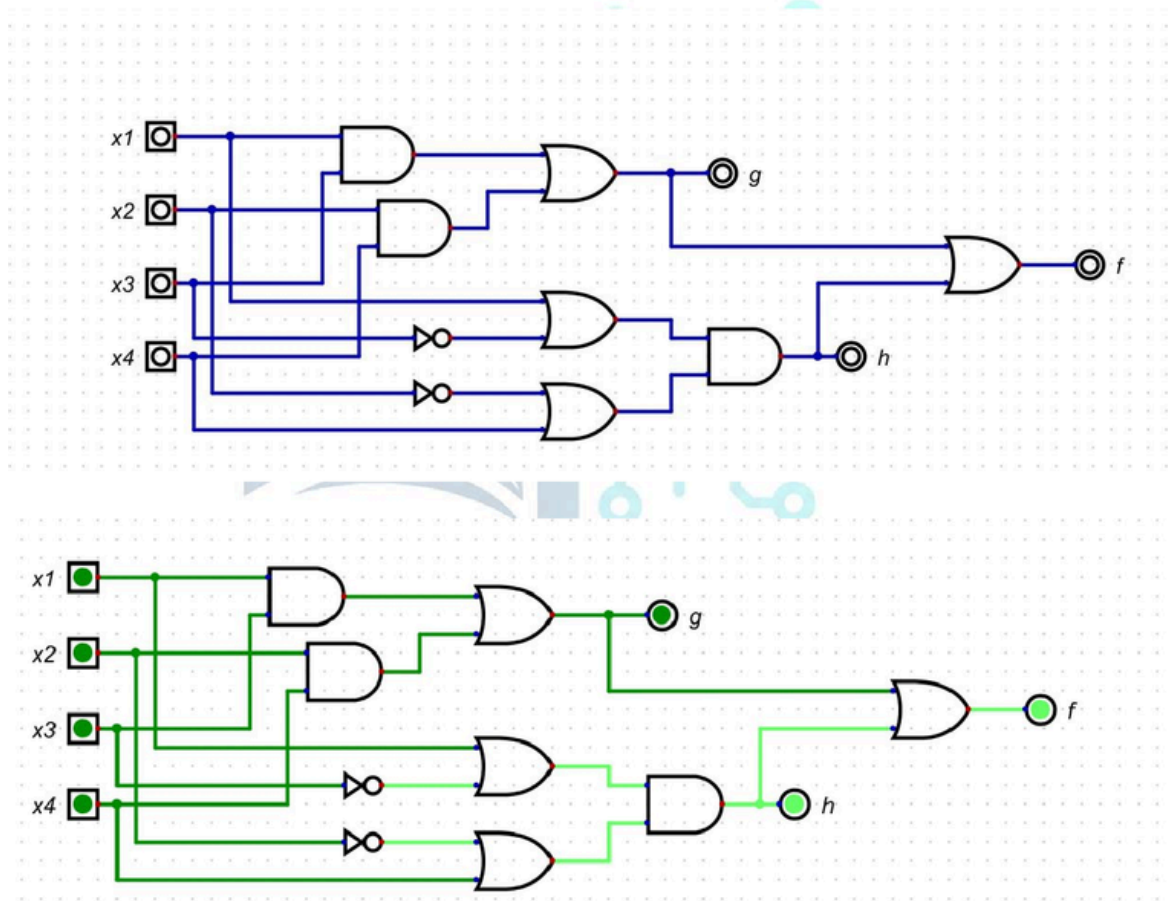
Task #3.1

```
module t1_3
  #(parameter n=3)
  (
    input logic [n-1 :0]x,
    output logic f
  );
  always_comb begin

    if((x[0]&x[1]) | (~x[1]&x[2])) begin
      f=1;
    end
    else
      f=0;
    end
  end
endmodule
```

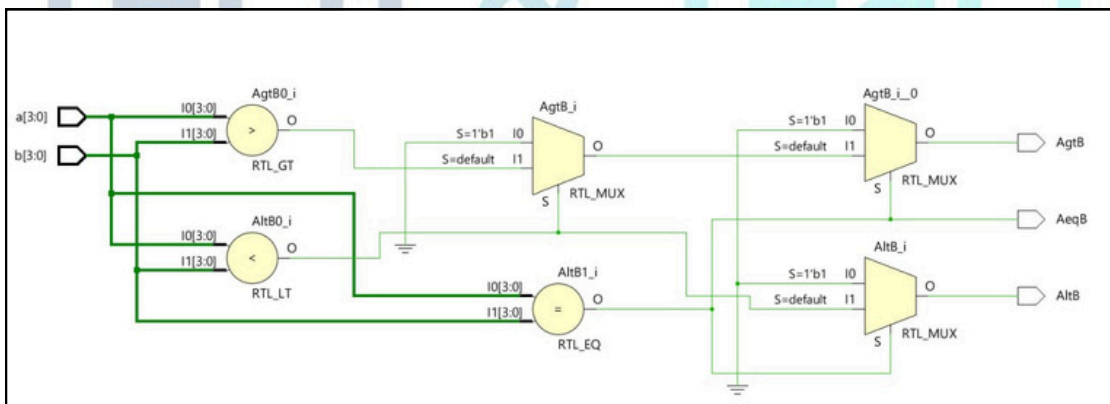


Task #3.2



Task #3.3

```
4 output logic AltB ,AgtB ,AeqB );
5 always_comb begin
6   if(a==b)begin
7     AltB=0;
8     AgtB =0;
9     AeqB=1;
10  end
11 else if(a<b)begin
12   AltB=1;
13   AgtB =0;
14   AeqB=0;
15 end
16
17 else if (a>b) begin
18   AltB=0;
19   AgtB =1;
20   AeqB=0;
21 end
22 else begin
23   AltB=0;
24   AgtB =0;
25   AeqB=0;
26 end
27 end
```



Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report				✓		
Task 3.1			✓		✓	✓
Task 3.2	✓	✓				
Task 3.3	✓	✓	✓	✓	✓	✓